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7. Throughput > 30 m²/h (equivalent to 1200 wafers per hour) for 20µm layer thickness. A simple scale-up is possible.

As per the current status of ProConCVD, all hardware installations have been completed. The transport and heating system is in operation. The infrastructure is online. The first high-quality epitaxial layers have been done successfully. Strategies to increase the efficiency of normal crystalline silicon solar cells include advanced metallisation, selective emitters, dielectric surface passivation, thinner wafers, process control, ultra-light trapping, material quality and back-contact cells. Estimating the efficiency potential on boron-doped Cz-Si, with a limitation due to metastable boron-oxygen defect, there is the optimised industrial cell structure (PERL). The efficiency is limited to about 20 per cent due to boron oxygen lifetime degradation. The solution: n-type silicon, with no degradation and higher tolerance to metal contamination.

The lab results of high-efficiency n-type PERL cells were also shared. There was substitution of local phosphorus diffusion by laser doping from innovative double-function PassDop layer (passivation and doping). Excellent results were achieved with evaporated front contacts.

Prof. Weber talked about the efficiencies of Ni/Cu/Sn metallisation. Solar cell properties include direct plating, lowly doped emitter (120 ohms/square) and dielectric rear passivation. It also has excellent efficiencies and fill factors. As for the thin-film CIS solar cell structure, the key challenge is to realise the impressive small-area lab efficiency results in production-size modules and volume production.

He touched upon the benefits of multi-junction solar cells and high-efficiency ISE triple-junction solar cells obtained by MOCVD thin-film deposition. Advantages of high-concentration PV cells include system efficiencies of 25 per cent AC today, about 200 MW/year worldwide production capacity, no cooling water or intentional hot water, modular kW to GW scale, and one-year energy payback time.

The future vision: Renewable electricity super grid

Prof. Weber gave an example of DESERTEC—the vision of an electricity super grid. DESERTEC is a mega renewable energy project that aims to set up a massive network of solar and wind farms stretching across the Middle East and North Africa (MENA) region and connected to Europe via a Euro-Mediterranean electricity network made up of high-voltage direct current transmission cables. The project, estimated at €400 billion, will provide 15 per cent of Europe's electricity by 2050. ●

The author is an executive editor at EFY